ECO-ROUTE

Data Quality Validation Report

Task 6 • NYC EV Mobility Data Warehouse • Phase 2

Report date: 14 November 2026
Database: nyc_ev_mobility_dw
Schema layers: project_staging (Bronze), project_dw (Silver), project_reporting (Gold)

Executive summary

The Phase 2 warehouse was evaluated using completeness, referential-integrity, reconciliation, numeric-validity, categorical-validity, and timestamp-consistency checks. The core fact-table relationships and three reconciliation totals matched. The checks also identified source-data exceptions: negative financial values, missing on-scene timestamps, and timestamp-order inconsistencies. These exceptions are documented for review and were not silently corrected or deleted.

Area	Result	Finding
Referential integrity	13 checks; 0 orphan records	All tested foreign-key relationships passed.
Completeness	18 selected fields; 0 NULLs	All selected mandatory/important fact fields passed.
Reconciliation	3 of 3 passed	Trip count, driver pay, and charger coverage matched.
Numeric validity	14 checks with zero negatives; 2 categories with exceptions	4 negative base fares and 2 negative driver payments.
Categorical validity	Y/N only	Both tested flags contain only expected values.
Timestamp validity	Exceptions identified	3,603 chronology exceptions; 133,087 missing on-scene timestamps.

Overall assessment: The warehouse passed the tested relationship and reconciliation checks. It is not accurate to describe every data-quality check as fully passed because the validation also found timestamp and financial exceptions that need source/business-rule review.

1. Scope and warehouse volumes

This report records the validation results supplied from the PostgreSQL project database. The checks cover selected important columns and relationships; they do not claim that every column in every table has been exhaustively profiled.

Table	Validated row count	Purpose
project_dw.fact_trips	500,000	Trip-level measures and dimensions.
project_dw.fact_zone_hourly_snapshot	197,160	Zone-hour aggregate snapshots (265 zones × 744 hours).
project_dw.fact_charger_coverage	26,846	Mapped charging-station coverage by date (866 stations × 31 dates).

Validation scope

The source-side timestamp and financial investigations used the cleaned staging view **project_staging.stg_trips_clean**. Fact-table checks used the corresponding tables in **project_dw**. The clean staging view standardizes/cleans records but the six negative financial values were already present there before fact loading.

2. Completeness checks

The completeness query checked 18 selected fields across the three fact tables. Every selected field returned zero NULL records.

Fact table	Selected fields checked	NULLs found
fact_trips	pickup_zone_key, dropoff_zone_key, company_key, date_key, time_key, weather_key, trip_miles, driver_pay	0 in all 8
fact_zone_hourly_snapshot	zone_key, date_key, time_key, weather_key, trip_count	0 in all 5
fact_charger_coverage	station_key, zone_key, date_key, num_ports, is_operational	0 in all 5
Total	18 selected fields	0

Result: PASS for the 18 selected fields. This result must not be generalized to every column. For example, wait_min was not part of these 18 selected fields, and a NULL wait_min was observed in one of the investigated trip records.

3. Referential integrity checks

Thirteen foreign-key relationships were checked. Each check counted fact rows whose non-NULL foreign key did not have a matching dimension key. All 13 checks returned zero orphan records.

Fact table	Relationships checked	Orphans
fact_trips	pickup_zone_key, dropoff_zone_key, company_key, date_key, time_key, weather_key (6)	0
fact_zone_hourly_snapshot	zone_key, date_key, time_key, weather_key (4)	0
fact_charger_coverage	station_key, zone_key, date_key (3)	0
Total	13 relationships	0

Result: PASS. All tested non-NULL foreign keys reference existing dimension records. NULL-key completeness was treated separately from orphan detection.

4. Reconciliation checks

Three aggregate reconciliations were performed between the detailed/aggregate fact data and the relevant expected totals.

Reconciliation	Source / expected	Compared value	Difference	Result
Trip count	COUNT(fact_trips) = 500,000	SUM(snapshot.trip_count) = 500,000	0	PASS
Driver pay	SUM(fact_trips.driver_pay) = 9,147,569.11	SUM(snapshot.total_driver_pay) = 9,147,569.11	0.00	PASS
Charger coverage	866 mapped stations × 31 dates = 26,846	fact_charger_coverage = 26,846	0	PASS

Result: 3/3 PASS. The trip count and driver-pay aggregate in the hourly snapshot reconcile exactly to the trip fact table. Charger coverage contains one row for each of 866 mapped stations across 31 dates.

5. Numeric validity checks

The numeric validation query searched for negative values in selected measures. Fourteen checks returned zero negative records. Two measures contained exceptions:

Measure checked	Negative records	Result / interpretation
trip_miles	0	PASS
trip duration (trip_time_sec)	0	PASS
base_fare	4	EXCEPTION — review negative fares
driver_pay	2	EXCEPTION — review negative driver payments
tolls	0	PASS
sales_tax	0	PASS
wait_min	0 negative	PASS for negative-value check; NULL completeness is separate
snapshot trip_count	0	PASS
snapshot total_wait_min	0	PASS
snapshot avg_trip_miles	0	PASS
snapshot rainfall_mm	0	PASS
charger num_ports	0	PASS
charger distance_to_zone_center_km	0	PASS
tips	0	PASS
congestion_surcharge	0	PASS
airport_fee	0	PASS

The negative base-fare and driver-pay counts do not overlap: 4 records had negative base_fare, 2 had negative driver_pay, and 0 had both negative. Therefore, six distinct trip records were identified for review.

Investigated financial records

Trip ID	Base fare	Driver pay	Other relevant observation
105537	-2.75	13.06	Congestion surcharge 2.75
151318	-2.75	5.39	Congestion surcharge 2.75
296774	-0.82	11.98	BCF 0.05; sales tax 0.15; airport fee 2.50
153080	-0.72	65.75	BCF 2.69; sales tax 8.68
357261	80.86	-6.94	Tolls 6.94; wait_min NULL
454637	108.16	-13.88	Tolls 13.88

All six corresponding records were also found in stg_trips_clean before fact-table loading. This shows that these values were present in the cleaned staging data and were not introduced by the fact-table load. The two negative driver_pay values exactly match the negative of their tolls values; this pattern may warrant business-rule investigation, but it does not by itself prove that the records are refunds, adjustments, or valid transactions.

6. Categorical validity checks

The distinct values and record counts for shared_request_flag and wav_request_flag were inspected.

Flag column	Value	Records
shared_request_flag	N	481,191
shared_request_flag	Y	18,809
wav_request_flag	N	498,869
wav_request_flag	Y	1,131

Result: PASS for observed categories. Both columns contain only Y and N in the 500,000 trip records. The displayed counts sum to 500,000 for each flag.

7. Timestamp completeness and chronology

Timestamp checks were performed on project_staging.stg_trips_clean. The chronology comparisons were counted separately from NULL checks because comparisons involving NULL do not evaluate to TRUE in PostgreSQL.

Timestamp completeness

Timestamp column	Missing records
request_datetime	0
on_scene_datetime	133,087
pickup_datetime	0
dropoff_datetime	0

All 133,087 records with at least one missing timestamp were missing on_scene_datetime. The other three timestamp columns had no missing values.

Timestamp chronology

Chronology rule	Records	Result
pickup_datetime > dropoff_datetime	0	PASS
request_datetime > on_scene_datetime	3,470	EXCEPTION
on_scene_datetime > pickup_datetime	133	EXCEPTION
Both request > on-scene AND on-scene > pickup	0	No overlap

The two chronology groups are disjoint. There are therefore **3,603 distinct chronology-exception records** (3,470 + 133). The pickup/drop-off ordering check passed for all records counted by that check.

Sample inspection

A sample of 10 records was inspected for the chronology exceptions. The first sample showed request timestamps later than on-scene timestamps, while pickup and drop-off remained in the expected order. This demonstrates why chronology exceptions should be reported separately rather than treating every affected trip as unusable.

Missing on-scene timestamps and chronology exceptions are separate findings. The missing count does not mean 133,087 invalid trips; it means the on-scene event time is unavailable for those rows. The project team should confirm the source dataset's rules for optional on-scene timestamps before deciding whether these are acceptable omissions or data defects.

8. Overall findings and recommendations

Finding	Recommended action
All 13 tested FK relationships have zero orphans.	Retain the current dimension-key mapping and keep the FK checks in the repeatable validation process.
All 18 selected completeness fields have zero NULLs.	Keep the current checks, but consider extending completeness profiling to other important fields and timestamp columns.
All 3 reconciliation totals match exactly.	Keep these totals as regression checks after future reloads/backfills.
4 negative base fares and 2 negative driver payments.	Compare with the source data dictionary and business rules; do not automatically replace with zero or delete.
Both categorical flags contain only Y/N.	Keep an allowed-value check so unexpected categories are detected in future batches.
133,087 missing on-scene timestamps.	Confirm whether on-scene time is optional/commonly absent in this source. Report it distinctly from invalid chronology.
3,603 distinct chronology exceptions.	Review source semantics and timestamp generation. Preserve original values until a correction rule is approved.

Conclusion

The warehouse demonstrates strong referential integrity and aggregate consistency for the tested relationships and totals. The main outstanding data-quality concerns are timestamp availability/order and six negative financial values. These findings should be retained as documented exceptions and investigated against source documentation or business rules before any data correction is applied.

9. Validation SQL file organization

Keep the validation SQL scripts in the Task 6 folder so the results can be reproduced. The existing four scripts are a good start. Add separate scripts for the timestamp and categorical checks, rather than leaving those queries only in chat or scattered across unrelated files.

Suggested file	What it should contain
01_completeness_checks.sql	The 18 selected-field NULL checks. Add timestamp NULL counts here only if you want all completeness checks together; otherwise keep them in 05.
02_referential_integrity.sql	All 13 foreign-key/orphan checks.
03_reconciliation_checks.sql	Trip count, driver-pay total, and charger-coverage reconciliations.
04_validity_checks.sql	Negative numeric-value checks, including base fare, driver pay, tips, surcharges, tolls, and other tested measures.
05_timestamp_checks.sql	Timestamp NULL counts, chronology counts, overlap check, and sample-record inspection query.
06_categorical_validity_checks.sql	Distinct values and counts for shared_request_flag and wav_request_flag; optionally add an explicit allowed-values test.

Suggested practice: add a short header to each SQL file describing its purpose, tables, and expected interpretation. Keep the queries that produced the reported results unchanged or clearly label any later edits. Save the query outputs or screenshots as evidence if your instructor requires proof of execution.

10. Limitations

This report is based on the query outputs provided during the validation session. It does not independently re-run the SQL against the database. Completeness was checked for 18 selected fact-table fields, not every column. The negative financial values and timestamp exceptions have not been adjudicated against an authoritative source data dictionary. No

records were deleted or modified as part of this validation.

End of report • Eco-Route Data Quality Validation • Task 6